

# Sarah M. Coleman, PhD

**Open to relocation nationwide** · Available to start within 30 days, at own expense

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## Summary

Bioprocess development engineer specializing in **upstream process development** and **fed-batch fermentation**. Designed and implemented a first-in-company **fed-batch bioreactor** process end to end, covering equipment selection, feed strategy, control parameters, **SOPs**, **batch records**, and KPIs, and led **tech transfer** to manufacturing. Background spans microbial strain and fermentation process development, multi-omics analysis, and statistical modeling, with upstream PD experience at **Bristol Myers Squibb** (biologics, **CHO cell culture**) and **Takeda** (vaccines, Vero cells). PhD in Chemical Engineering and MS in Statistics, with experience in **scale-up**, **DoE**, and statistical analysis of experimental biology data in Python and R.

## Work Experience

**BioReNuva**, Austin, TX

*Bioprocess Development Engineer* (Contract/Fixed-term) | Jun 2026 – Present

- Established fed-batch fermentation capability for glycolipid production using *Pseudomonas* strains as the company's subject-matter expert, building on prior collaboration with the Alper Lab
- Designed and implemented the company's first fed-batch bioreactor process from scratch, including equipment selection, feed strategy, control parameters, and run protocols
- Authored the standard operating procedure (SOP) for bioreactor runs, establishing the reference process documentation used for ongoing and future fermentation campaigns
- Defined key performance indicators (KPIs) and identified process bottlenecks for glycolipid synthesis, informing the roadmap for scale-up and continuous improvement
- Led cross-functional technology transfer, translating process knowledge into documentation and protocols to hand off from development to manufacturing teams

**The University of Texas at Austin**, Austin, TX

*Postdoctoral Research Fellow, Statistics and Data Sciences* | Oct 2024 – Sep 2026

- Developed epidemiological and statistical models of environmental-health data in Dr. Roger D. Peng's group, including analyses across 5 Texas cities and dimension reduction of emission mixtures across approximately 3,000 U.S. counties, resulting in a first-author paper in *Environmental Research Communications* and a second under review at *Environmental Health*; advised medical faculty on statistical methodology and experimental design

**The University of Texas at Austin**, Austin, TX

*Lecturer, Statistics and Data Sciences* | Jan 2026 – Apr 2026

- Instructor of record for 100-student upper-division data science and machine learning course, rated 4.32/5 overall by the students, a higher rating than the department, college, and university averages; managed 3 TAs

**The University of Texas at Austin**, Austin, TX

*Graduate Student Fellow, Chemical Engineering* | Aug 2019 – Aug 2024

- Integrated CFD modeling with fermentation experiments to characterize bioreactor performance at up to 50% (v/v) waste cooking oil in 3 L bioreactors, among the highest loadings reported, and simulated scale-up to 4,000 L
- Evolved *Yarrowia lipolytica* to tolerate up to 30% (v/v) hydrothermal liquefaction wastewater (up from 10%), yielding an isolate with nearly 3-fold higher titer of the high-value polyketide triacetic acid lactone while reducing COD (chemical oxygen demand)
- Evolved *Y. lipolytica* for growth on high-density waste cooking oil and engineered it to produce long-chain fatty alcohols, characterizing the evolved strain by genomics, transcriptomics, and metabolomics
- Developed a genome editing system using Golden Gate assembly for targeted engineering of *Y. lipolytica*, and used this system to screen a native promoter library
- Proposed and validated a linear mixed-effects model to quantitatively evaluate the performance of biological Boolean logic gates for the first time, published in *ACS Synthetic Biology*
- Mentored 4 undergraduate researchers and 2 junior PhD students, and presented results to project sponsors and collaborators

**MathWorks**, Remote (Natick, MA)

*MATLAB Student Ambassador* | Mar 2020 – Jul 2021

- Taught technical workshops on MATLAB-based modeling and data analysis workflows for academic researchers

## **Massachusetts Institute of Technology, Cambridge, MA**

*Advanced Undergraduate Researcher* | Sep 2018 – Aug 2019

- Developed a method of nanophosphor size sorting in Dr. Michael S. Strano's lab to improve the capacitance of nanoparticles infiltrated into a light emitting plant

## **Massachusetts Institute of Technology, Cambridge, MA**

*Undergraduate Researcher* | Sep 2016 – Aug 2018

- Developed anaerobic assays in Dr. Gregory Stephanopoulos's lab to characterize the carbon assimilation of *Moorella thermoacetica* enzymes in the Wood-Ljungdahl pathway

## **Bristol Myers Squibb, Devens, MA**

*Intern, Upstream Process Development (Biologics)* | Jun 2018 – Aug 2018

- Designed and executed metabolite-controlled fed-batch cell culture experiments in 12 parallel 5 L bioreactors to systematically evaluate monoclonal antibody production performance in CHO cell lines
- Analyzed upstream bioprocess data to identify conditions that improved yield stability and process robustness in biologics production

## **Takeda Pharmaceuticals, Cambridge, MA**

*Intern, Process Development (Vaccines)* | May 2017 – Aug 2017

- Optimized Vero cell culture conditions to improve vaccine production performance within a next-generation manufacturing system
- Ran tangential flow filtration (TFF) for vaccine manufacturing, and helped design experiments to improve process scalability and production efficiency
- Performed analytical assays to determine infectious vaccine titer (CFU)

## **Education**

### **The University of Texas at Austin, Austin, TX**

*Ph.D. in Chemical Engineering*, Aug 2024 (Advisor: Dr. Hal S. Alper)

### **The University of Texas at Austin, Austin, TX**

*M.S. in Statistics*, May 2024

### **Massachusetts Institute of Technology, Cambridge, MA**

*S.B. in Chemical-Biological Engineering*, Jun 2019 (Research advisors: Dr. Michael S. Strano, Dr. Gregory Stephanopoulos)

## **Technical Skills**

- **Bioprocessing & Fermentation:** Batch and fed-batch microbial fermentation, bioreactor operation with DO and pH control (BioFlo/CelliGen, DeltaV), high-throughput growth screening (BioLector, 96-well plates), scale-up, kLa and mass transfer, media and feed development, seed train expansion, tangential flow filtration (TFF), SOPs, batch records, tech transfer
- **Synthetic Biology & Metabolic Engineering:** Metabolic engineering, genome editing (Golden Gate assembly, CRISPR), plasmid design and cloning (*E. coli*), strain engineering, promoter library screening, adaptive laboratory evolution
- **Cell Culture & Molecular Biology:** Yeast, bacterial, and mammalian (CHO, Vero) cell culture, aseptic technique, cGMP, Vi-CELL, Cedex Bio, NucleoCounter, flow cytometry, qPCR, protein purification, Western blotting
- **Computational Biology & Data Science:** Python, R, SQL, MATLAB, JMP, statistical modeling, machine learning, data visualization, DoE, CFD (Ansys CFX)
- **Bioinformatics & Omics:** RNA-seq and Tag-seq (DESeq2), genomic variant analysis, genome annotation, metabolomics, high-performance computing, Docker
- **Analytical Chemistry:** HPLC, LC-MS/MS, GC-MS, headspace GC, UV-Vis, assay development, plate reader assays
- **Software:** Benchling, LIMS, Linux, Bash, Git/GitHub, Jupyter, LaTeX, GraphPad Prism

## **Select Honors and Awards**

- **First Place, Professional Division at the Data Challenge Expo, JSM (Joint Statistical Meetings)**, 2025
- **National Science Foundation Graduate Research Fellowship (NSF GRFP)**, 2021
- **Provost's Graduate Excellence Fellowship**, UT Austin, 2019
- **Individual Accomplishment Citation, MIT Department of Chemical Engineering**, 2018 & 2019  
Only known undergraduate to win this award two years in a row, for volunteer work and service to the department

## **Publications**

14 peer-reviewed publications, seven as first or co-first author (\* co-first). Selected:

- **Coleman, S. M.\*** et al. High oil density bioreactor-scale fermentations of *Yarrowia lipolytica* using CFD modeling and experimental validation. *Biotechnology Journal* (2024)
- **Coleman, S. M.\*** et al. Evolving tolerance of *Yarrowia lipolytica* to hydrothermal liquefaction aqueous phase waste. *Applied Microbiology and Biotechnology* (2023)

Full list: [sarahmcoleman.github.io/papers.html](https://sarahmcoleman.github.io/papers.html)